TILITER Retail

Tiliter Vision System Setup and Integration Guide

for POS software providers and retailers

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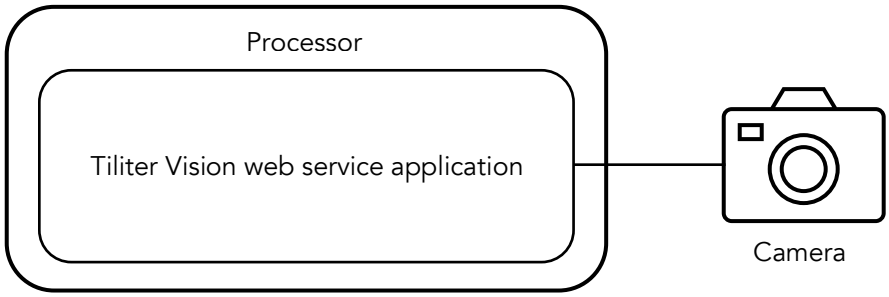
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Background

The Tiliter Vision system is a pre-trained computer vision system that can identify fresh produce such as fruit, vegetables, baked goods, nuts etc. It can capture an image and make a prediction within a fraction of a second.

The Tiliter Vision system includes the subscription-based SaaS Tiliter Vision application that runs as a web service, a camera, and a processor.



The Tiliter Vision system can be integrated into a point of sale (POS), self-serve checkout (SCO), retail scale or any other hardware system that can benefit from product recognition.

Scope

This document starts by showing you an overview of the user experience process, followed by examples of how the result looks to a shopper on the user interface.

It then takes developers (both Tiliter customers and team members) through the steps to integrate with Tiliter's software. It explains how to initiate a prediction, and how to interpret the response. It also includes links to all other supporting endpoints. This document covers Tiliter Vision software running on a Tiliter Vision processor retrofitted into an existing POS system, or designed and built into an OEM solution.

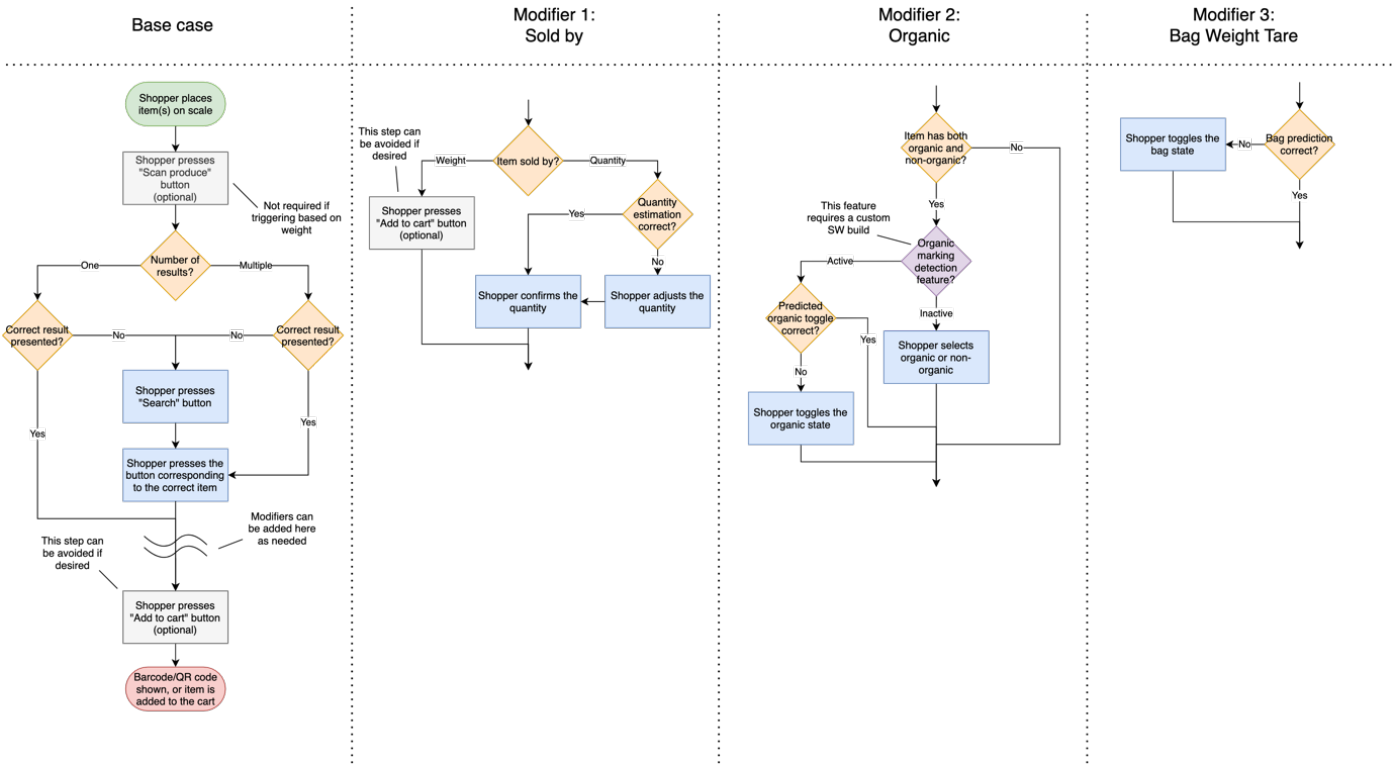
Section 1

The user experience

Process flowchart

The examples that follow deep dive into specific scenarios. These fall within a process workflow, shown here:

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The modifiers have been split out in the flowchart to make it easier to read. These events are only triggered under certain circumstances and can be input into the shopper workflow as needed.

Understanding the user experience reference design

To start the process of integrating the Tiliter Vision system into a POS, SCO, or other existing system, we'll take a look at example reference designs. These designs will give you a better idea of the look and feel that should be presented to shoppers. The goal is to make the experience simple and easy to use.

Example Shopper Workflows – Base Cases

First, we'll examine the three primary outcomes that shoppers typically experience.

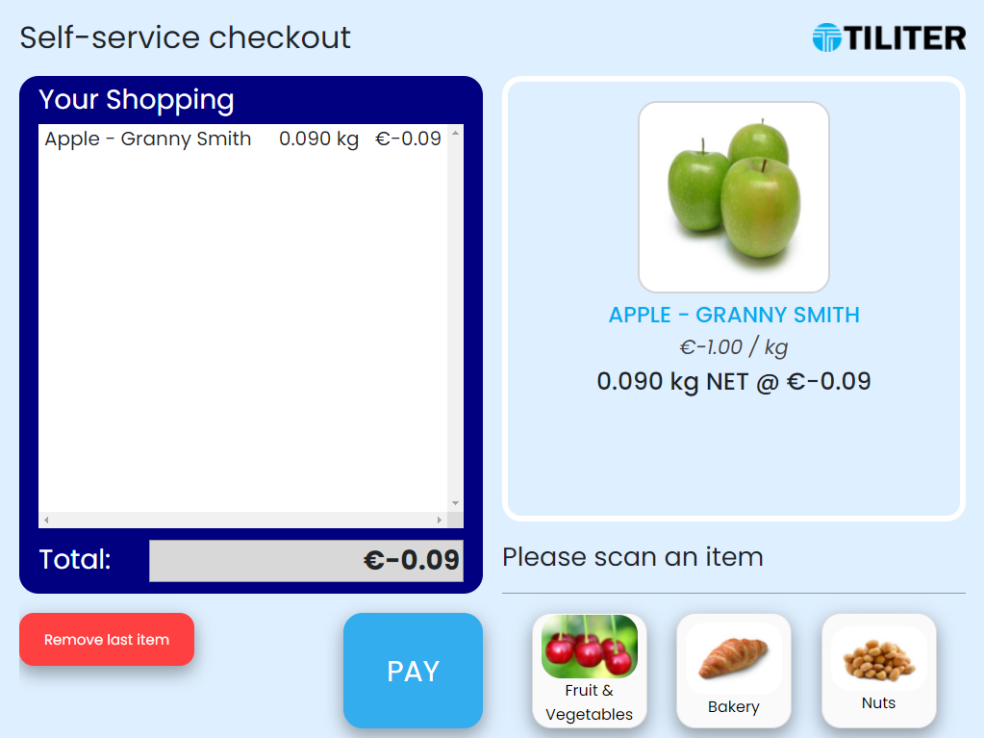
Example 1: One result, correct prediction

This is the best outcome possible. In this case, the shopper places the item down and the only result shown is the correct one. This enables the POS to automatically add the item to the cart if desired.

This occurs approximately 40-90% of the time. Upon request, Tiliter can tune the software towards either end of this spectrum. The more aggressive the tuning the greater chance that an incorrect result will be the only choice.



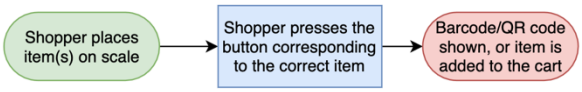
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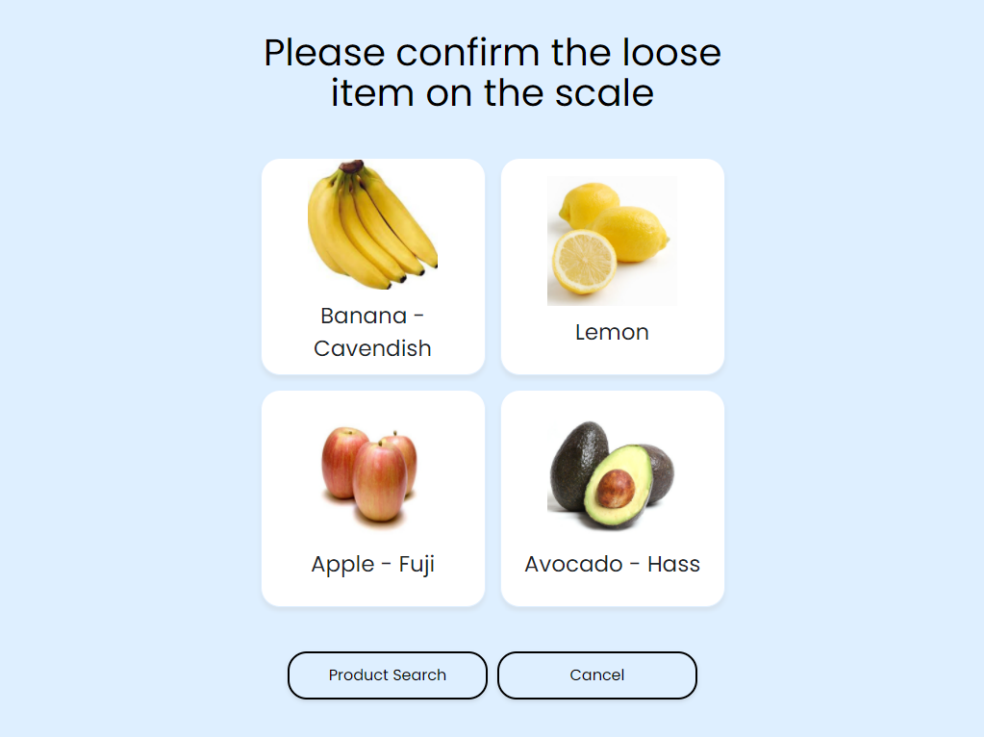
Example 2: Multiple results, correct prediction

In this case, several categories are predicted and one of them is correct. The shopper will need to make a simple selection between a small number (2-4 for instance).

This outcome, along with the above, occur approximately 95-99% for a mature system with all categories supported.



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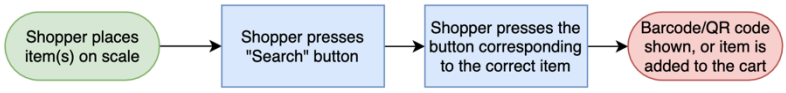


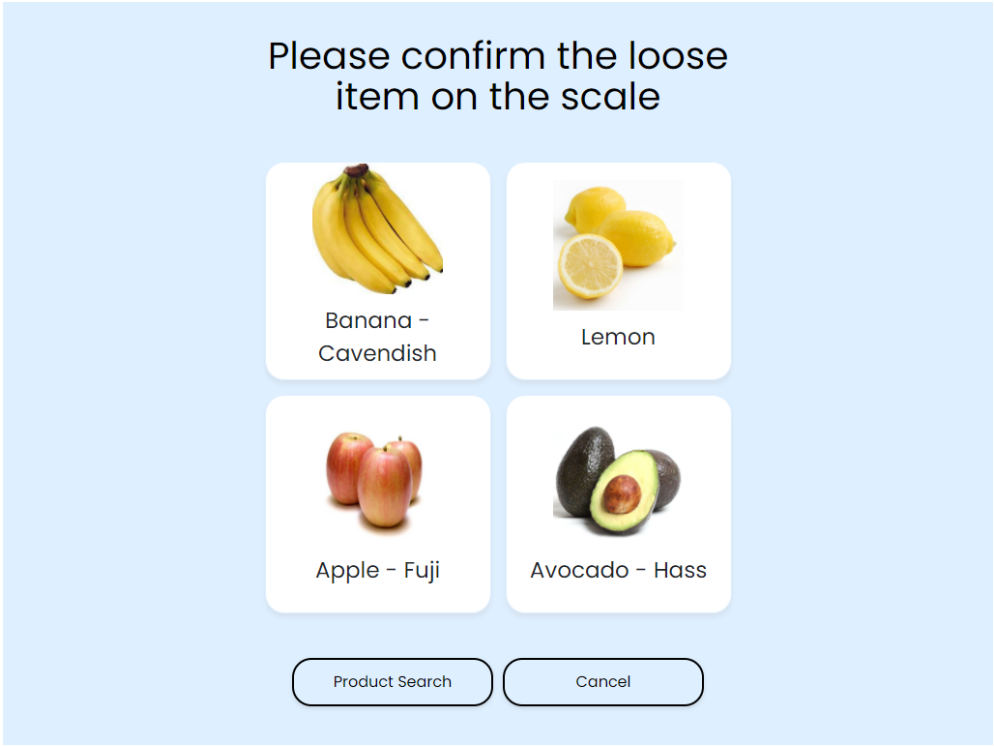
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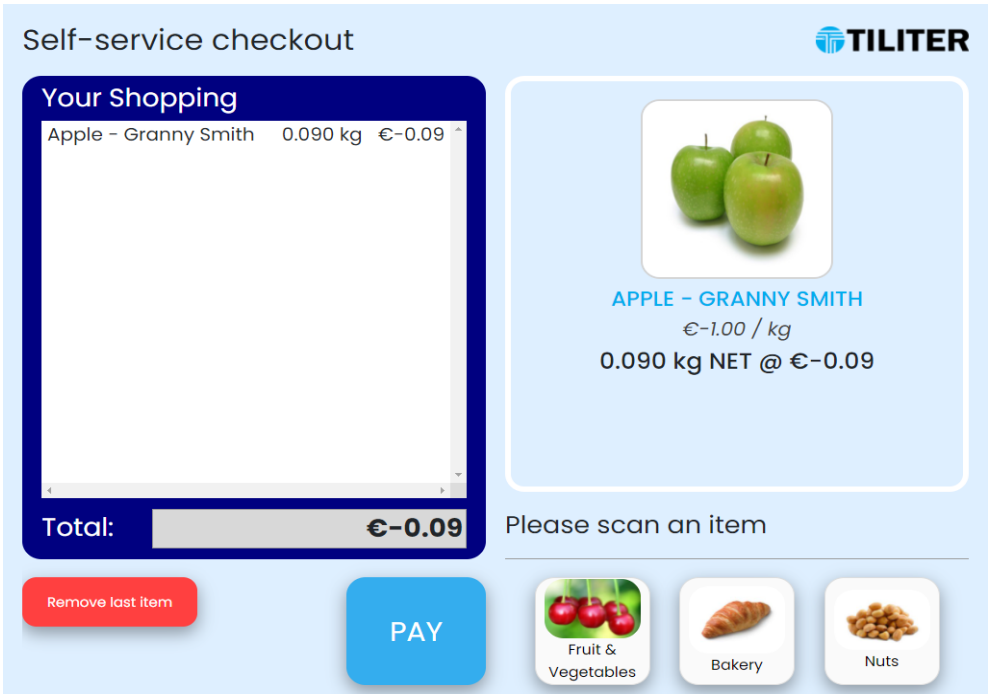
Example 3: Incorrect prediction

When all the predictions made are incorrect, the shopper must manually search for the correct item.

This occurs approximately 1-5% of the time.







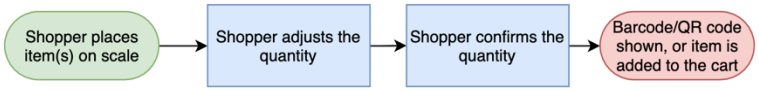
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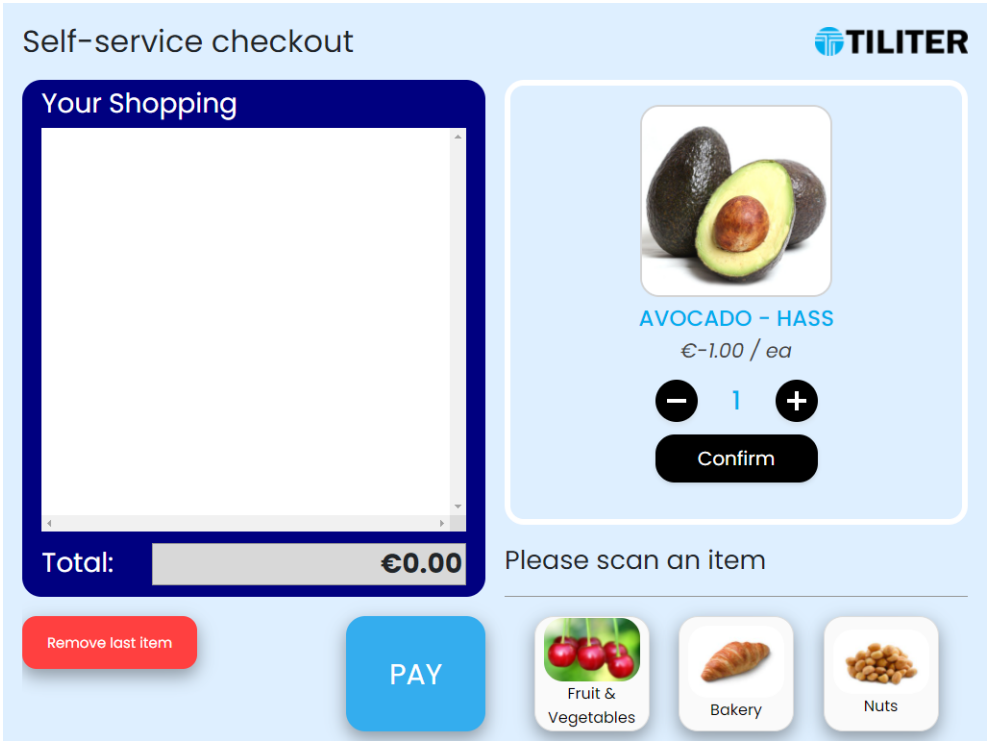
Example Shopper Workflows – Additional Cases

In some cases, various features are active and have gone wrong. This only occurs in a minority of cases — usually, the workflow defaults back to the base cases, shown above.

Example 4: One result, correct prediction, incorrect quantity estimate

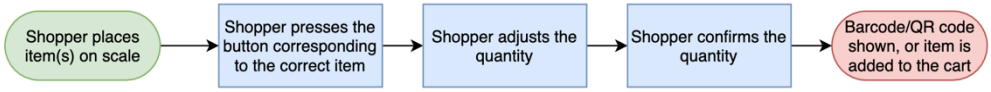
In this case the shopper needs to override the quantity estimation, as the value shown is incorrect. The shopper can adjust and then confirm the quantity, using the interactive update quantity button onscreen.



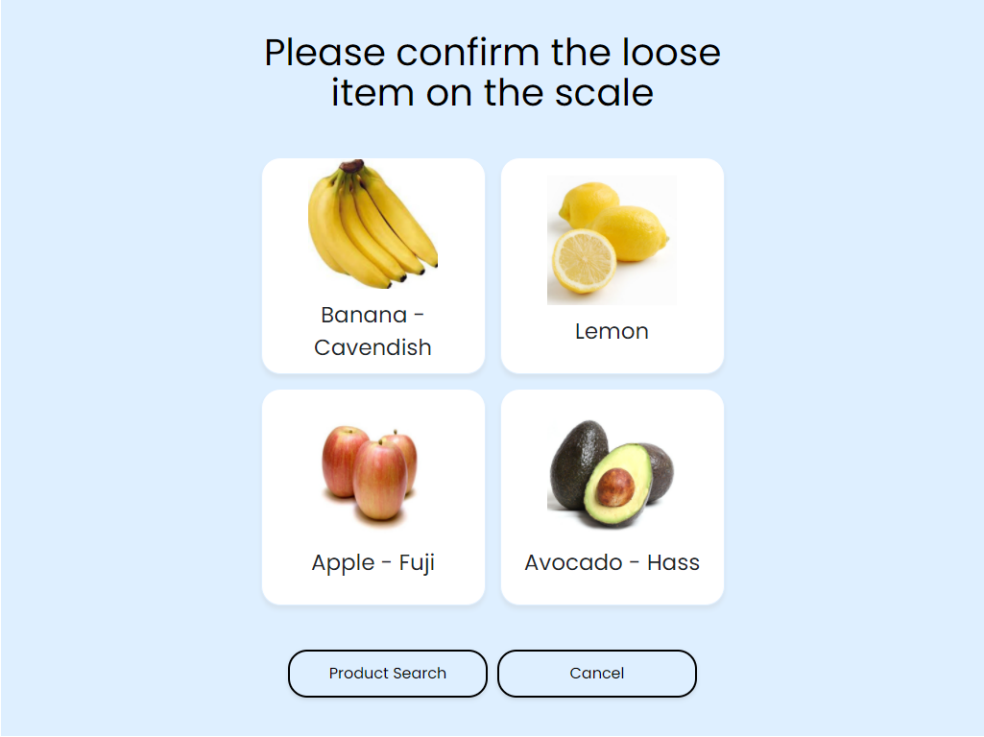


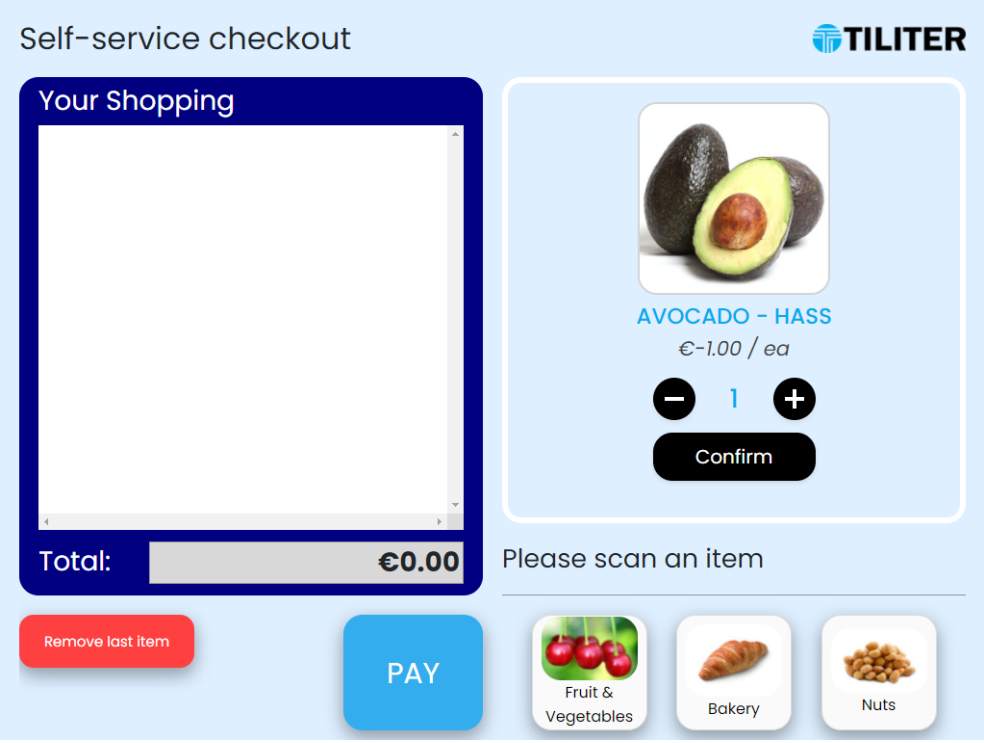
Example 5: Multiple results, correct prediction, incorrect quantity estimate

This is a combination of examples 2 and 4, where the Tiliter Vision system presents multiple options, but the quantity estimate is incorrect.

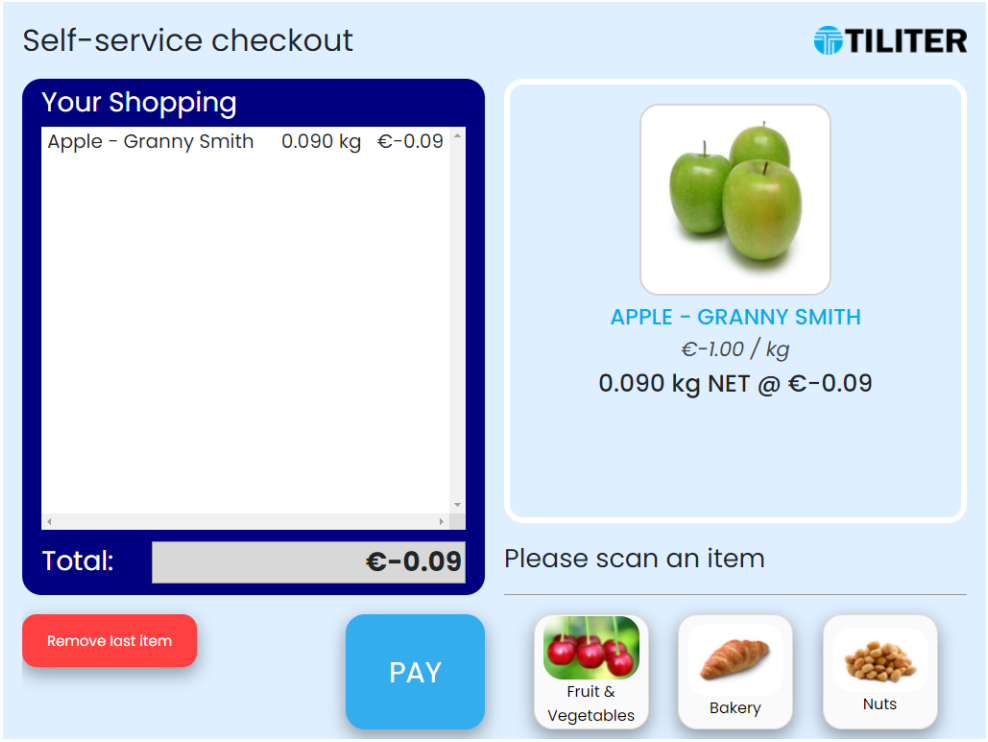


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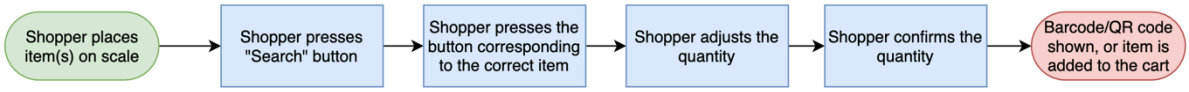


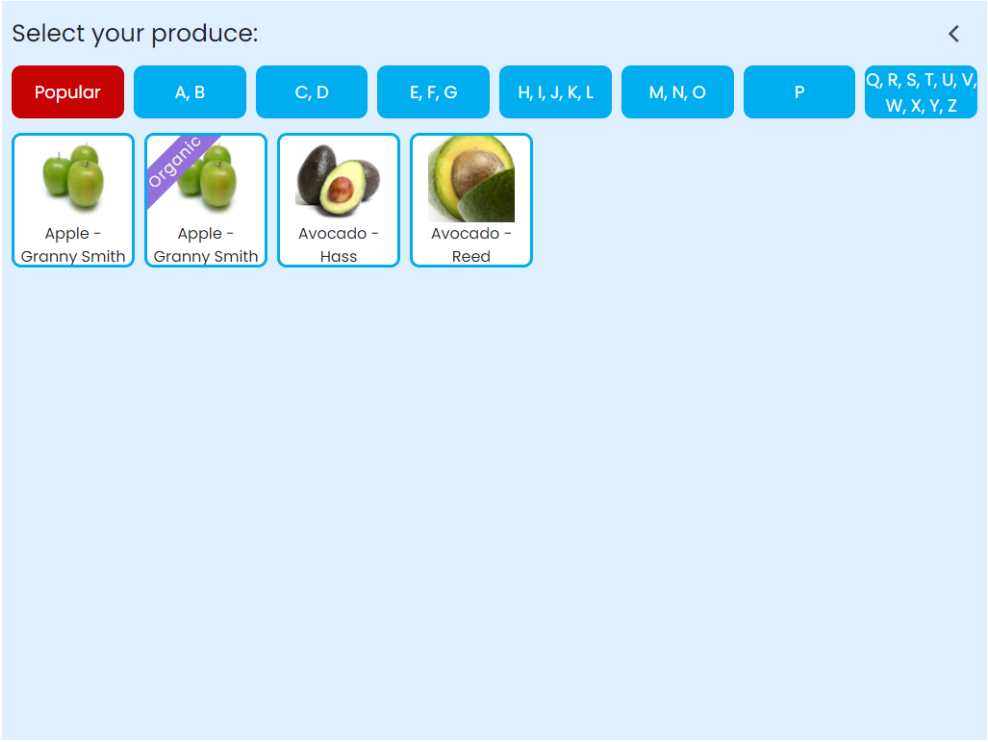


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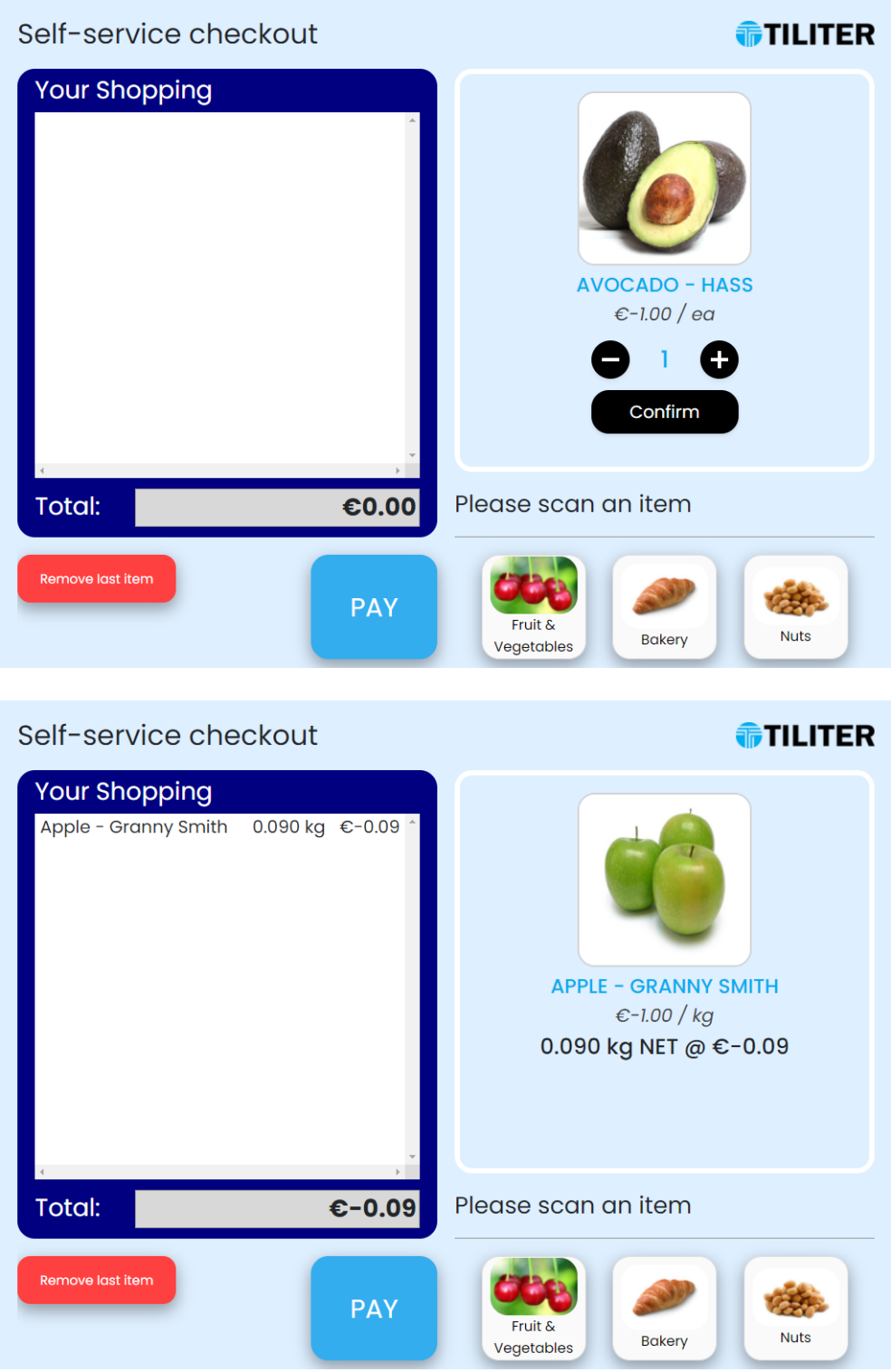


Example 6: Incorrect prediction, incorrect quantity estimate

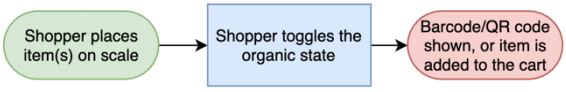




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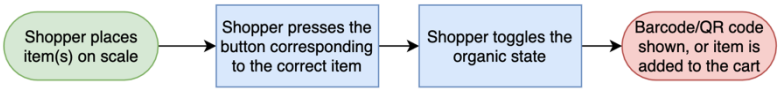


Example 7: One result, correct prediction, incorrect organic prediction

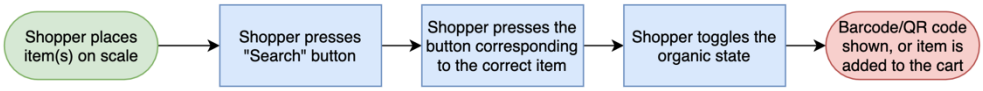


Example 8: Multiple results, correct prediction, incorrect organic prediction

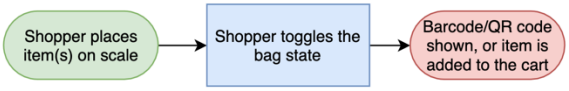
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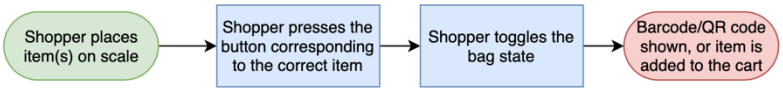
Example 9: Incorrect prediction, incorrect organic prediction



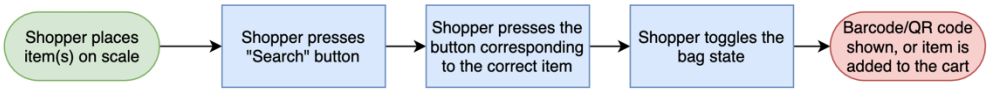
Example 10: One result, correct prediction, incorrect bag prediction



Example 11: Multiple results, correct prediction, incorrect bag prediction



Example 12: Incorrect prediction, incorrect bag prediction

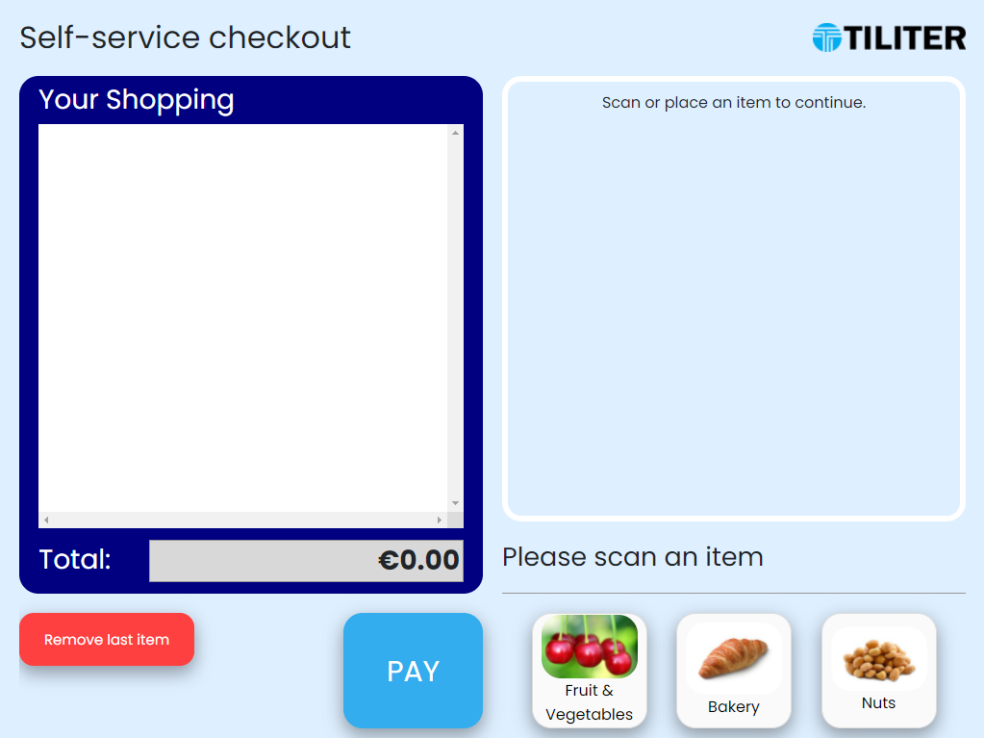


How to understand the user interface

In this section we'll look at the screens that the shopper is presented with, and the important features on each screen. This will help you set up your own user interface in line with our recommended best practice.

Standby screen

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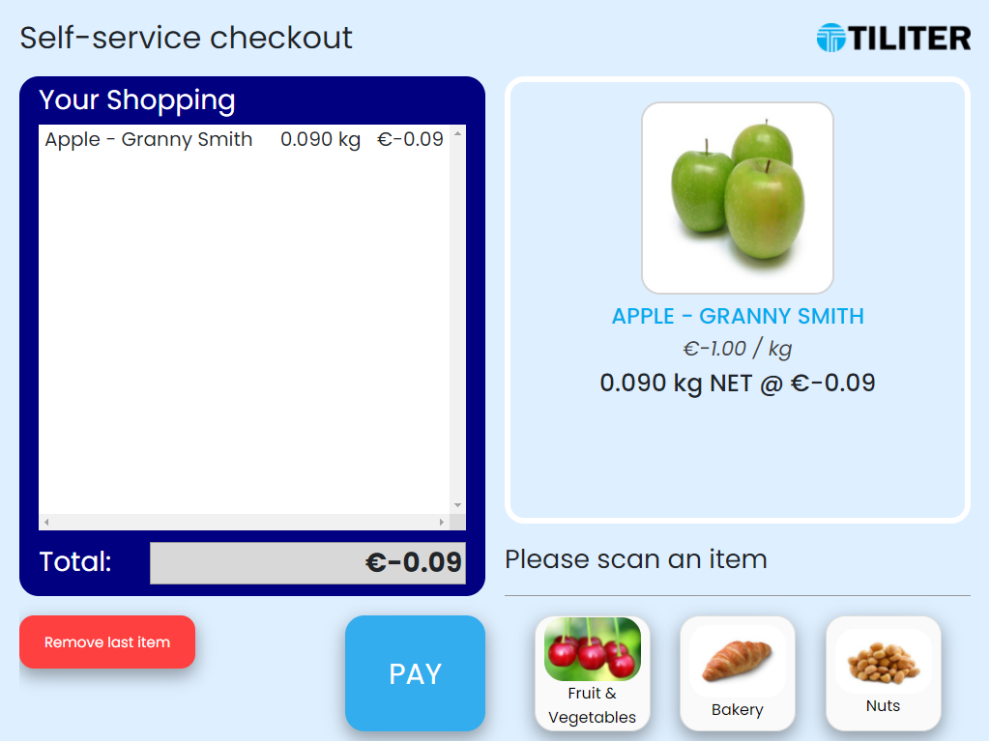
What's important on this screen?

- It's friendly, welcoming, well-designed, simple and modern
- UI elements are centred with the correct aspect ratio and high attention to detail
- It's obvious for the shopper what to do next:
    - If the scale weight being stable is the trigger, the screen should clearly communicate that an item should be placed down to begin
    - If an on-screen button is the trigger, it should be obvious which button to press to continue – ideally the only button on the screen
- When the button is pressed (or scale weight is stable), a prediction request should be sent to the Tiliter Vision system

Prediction – One result

If there's only one prediction result, here's an example of the screen shown to the shopper:

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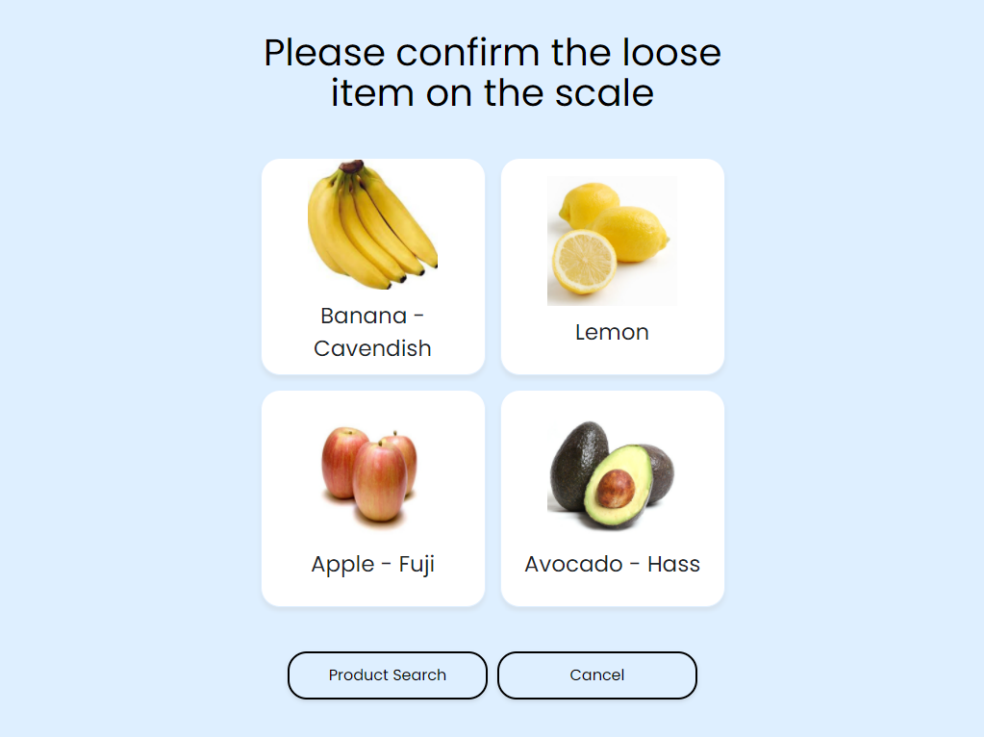
What's important on this screen?

- The selected category is clearly shown and dominates the screen
- The category has a clear, high resolution image that's set to the correct aspect ratio
- The shopper has the ability to override the prediction

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Prediction – Multiple results

This is an example of the screen shown to the shopper if there are multiple predictions:



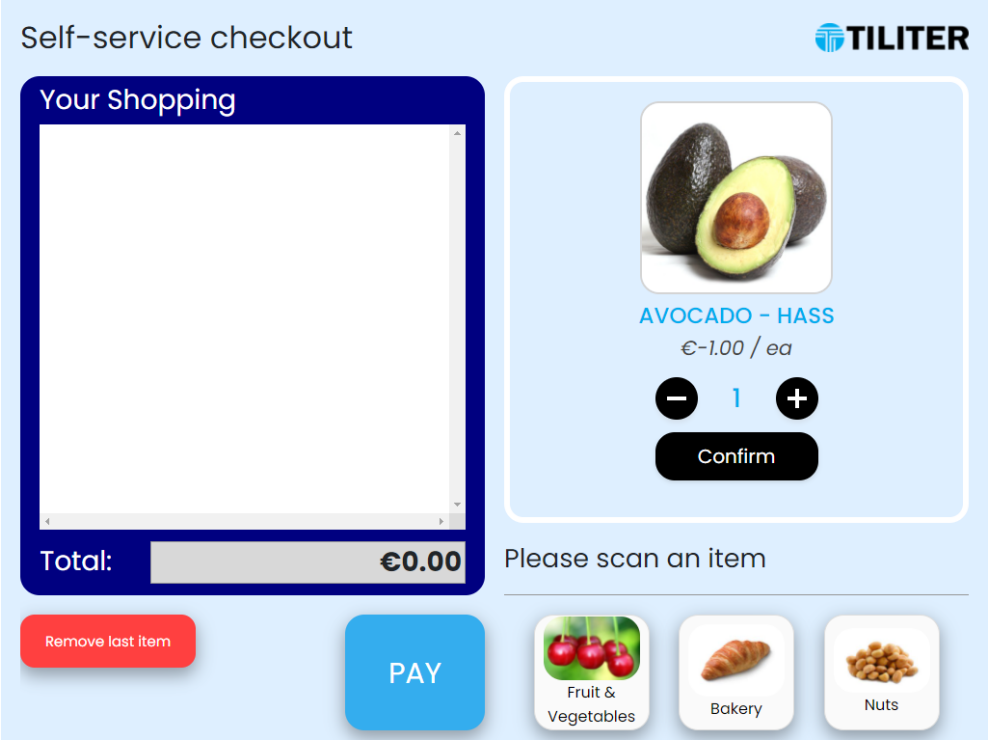
What's important on this screen?

- All predictions made by the Tiliter Vision system are presented to the shopper
- The options are prominent and presented in the centre of the screen
- If the correct option is not presented, there's an obvious and easy-to-access button to take the shopper to the search screen

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Quantity adjustment

If the shopper places more than one item on the scale, the screen will look like this:

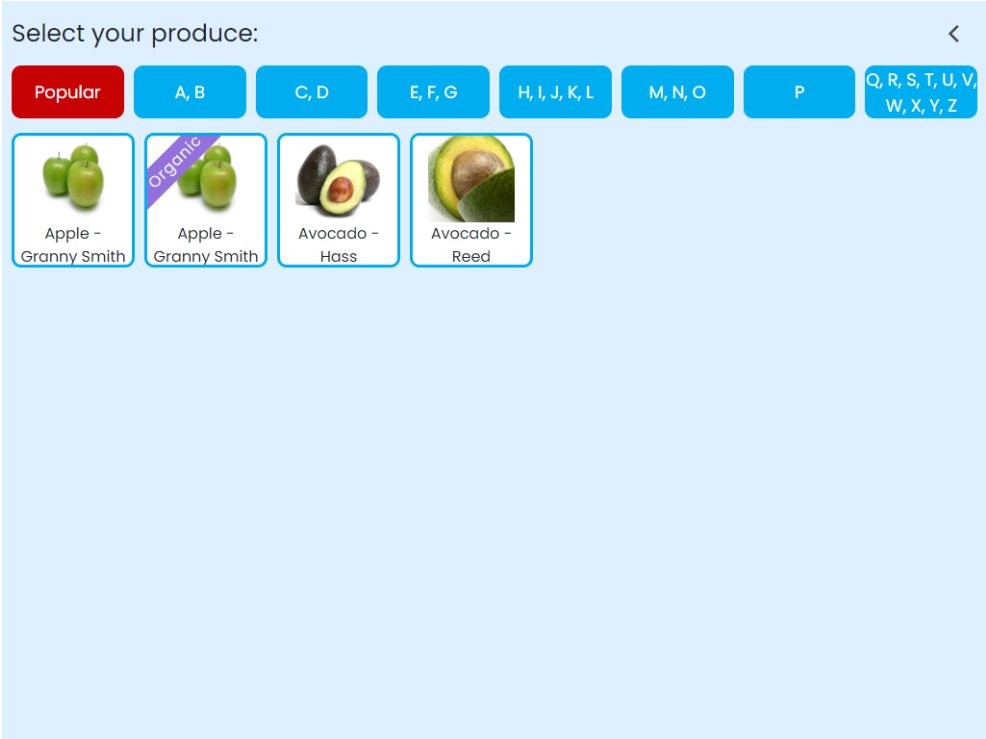


What's important on this screen?

- There is a prominent quantity adjustment mechanism (plus and minus buttons) that allows for the shopper to adjust and re-adjust as needed
- The first quantity is estimated based on the measured weight divided by the average weight
- Otherwise, the screen looks identical to the one prediction result screen

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Category search





Important aspects of this screen:

- There are broad groups presented at the top
- There's an option to view popular categories that allow you to see subgroups e.g., all Apples, all Mandarins, all Mangos

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- All options are presented with a uniform size and are large enough for shoppers to easily read the writing and recognise the images

Special category handling

There are a few special cases that have their own screen. These are:

- A hand is detected
- A phone is detected
- Non-produce items are detected
- Background is detected

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Section 2

How to integrate Tiliter software and hardware with a POS unit

This section explains the steps you need to follow to integrate a POS unit with the Tiliter Vision system. The target end state should be as described in Section 1 above.

There are two key steps to integration:

1. **Hardware integration** — Physically connecting the hardware and setting up reliable communications between the TV processor and the POS unit.
2. **Software integration** — Updating the POS unit's software revision to allow for Tiliter Vision predictions

How Tiliter communicates with a POS

The Tiliter Vision processor hosts a web service. This web service allows any connected devices to run predictions and access endpoints – the communication points at either end.

These endpoints are referenced in this document:

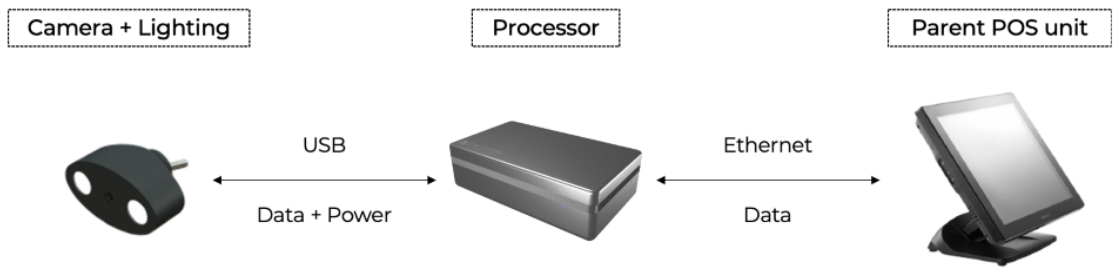
1. backend/*result*
2. backend/*get_weight*
3. backend/*label_last_result*
4. */activelist*

You can find a full list of our endpoints here: Postman documentation.

All interactions with the Tiliter Vision processor are done through these endpoints.

How to connect to the Tiliter Vision processor

First, set up your Tiliter hardware. Below is an example of a locally run Tiliter Vision POS integration setup. In this case, a microcontroller is used to scan and identify the item presented. For more details, please refer to the **Hardware setup guide.**



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After you've physically connected the device, the POS needs to know how to identify the Tiliter Vision processor that's mounted to it. You can do this in a couple of ways.

Option 1: Use a zeroconf network tool and the Tiliter Vision hostname

The simplest way to directly connect to a Tiliter Vision processor is to install a zeroconf network tool such as Bonjour and address the processor by its hostname. You can find the hostname on the device, near the serial number. If you use this option, the device hostname will be used to address it.

Option 2: Allocate fixed IP based on MAC address using DHCP

If the Tiliter Vision processor is connected via a network device, such as a switch or router, that device can assign a fixed IP to the processor based on its MAC address. You won't need to install Bonjour. If you use this option, the fixed IP will be used to address the device.

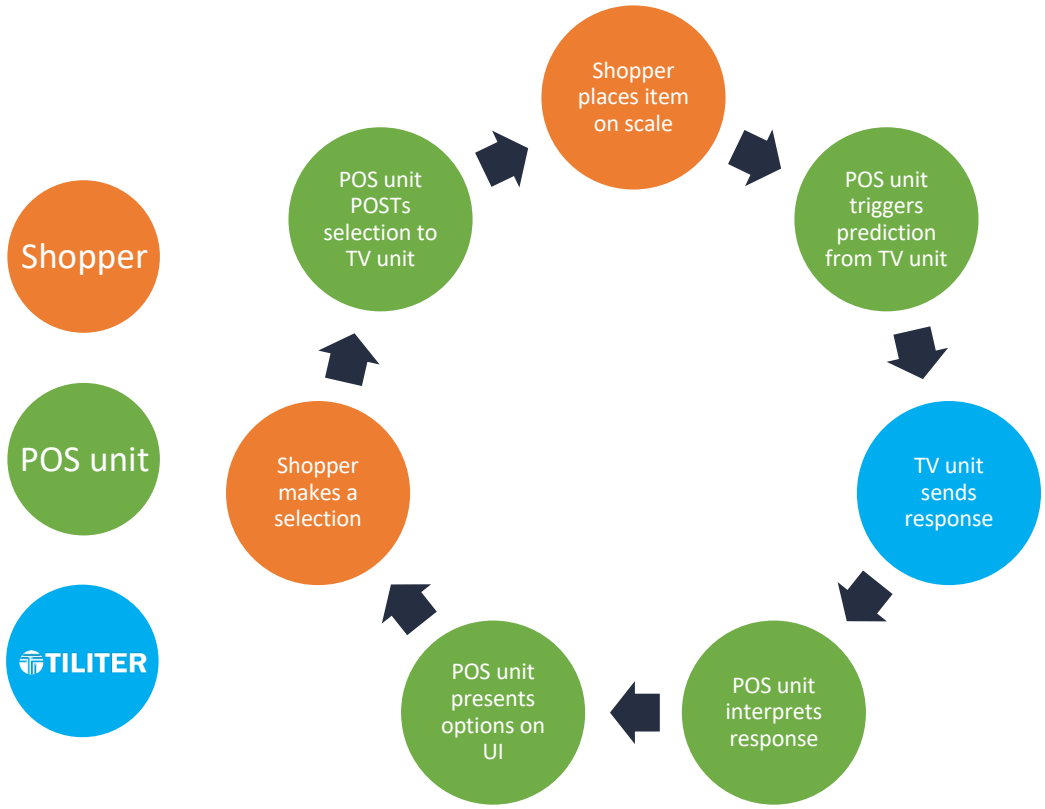
Understanding the Tiliter software workflow

Each time the Tiliter software is used, the following workflow takes place:

1. Shopper requests a prediction from the POS.
2. POS triggers a prediction from the */result* endpoint on the Tiliter Vision processor.
3. Tiliter Vision processor sends POS a JSON string with all prediction information.
4. POS interprets JSON string.
5. Decision point:
    a. If only one category is in the JSON string, item is added to cart.
    b. If two or more categories are in the JSON strong, POS presents shopper with selection, ranked based on score.
6. Shopper makes a selection if presented with the choice.
7. POS POSTs what was added to the cart to the */label_last_result* endpoint.

Here's a diagram of the workflow:

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These steps are described in the following sections in further detail.

The shopper places an item on the scale, triggering a prediction

From the shopper's perspective, the entire process should be seamless and intuitive. There are two ways a shopper can trigger a prediction:

Option 1: Scale weight stable

To get the most reliable results, we suggest that you set up the POS or connecting device to ask the HTTP server for a result once the scale weight is stable. This can be directly read from some scales if they provide an output for scale weight stability. Alternatively, some scales may only provide a reading when they are stable – so a transition from 0 to >0 would be sufficient.

As each scale is different it is recommended to consult with the scale manufacturer to decide the best way to identify when there is a stable weight reading.

Option 2: On-screen button

Another way to trigger a prediction is to provide an on-screen button.

If this is the preferred option, the integration is simple — the POS unit presents a button on the screen, and the shopper presses this button to request a prediction.

The POS unit triggers a prediction

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Once the POS unit is aware that the shopper is requesting a prediction, it must trigger a prediction on the Tiliter Vision system.

To trigger a prediction, set up the POS device to call the *backend/result* endpoint on the Tiliter Vision processor, and a response will be received a fraction of a second later. This endpoint can be found here:

http://<DEVICE IP ADDRESS OR HOSTNAME>:8000/backend/result

Examples:

- For software running on an external device with IP Address 169.254.240.146 the address is http://169.254.240.146:8000/backend/result/
- For software running on an external device with hostname TV123, the address is http://TV123:8000/backend/result/
- For software running on the host machine the address is http://localhost:8000/backend/result/

The Tiliter Vision units sends a response

Tiliter system predictions are formatted and stored in a JavaScript Object Notation (JSON) file. This makes it easy to integrate into most programming languages.

Understanding the JSON response

Here's an example of a typical JSON response:


```
{
  "guid": "ee5e30fd-65e0-4c02-8b3d-d52ec1f1ddf8",
  "weight_kg": 0.28,
  "prediction_time": 0.7546603679656982,
  "bag_predictions": [
    {
      "bag_name": "none",
      "score": 15.23
    },
    {
      "bag_name": "bakery",
      "score": 0
    },
    {
      "bag_name": "net",
      "score": 0
    },
    {
      "bag_name": "plastic",
      "score": 84.77
    }
  ],
  "organic_info": {
    "accuracy": 1.57,
    "threshold": 30
  },
  "predictions": [
    {
      "category": "Plum",
      "variety_1": "Red",
      "variety_2": null,
      "full_name": "Plum - Red",
      "custom_name": "Red Plum",
```


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```
      "custom_id": "691911",
      "custom_id_organic": "130586",
      "unit": "kg",
      "price": -1,
      "type": "product",
      "product_type": "fruit_and_vegetable",
      "tiliter_id": "21324b9e-a35c-48d1-89de-c3036607422a",
      "score": 84.98
    },
    {
      "category": "Plum",
      "variety_1": "Garnett",
      "variety_2": null,
      "full_name": "Plum - Garnett",
      "custom_name": "Plum - QG Antioxidant",
      "custom_id": "838820",
      "custom_id_organic": "0",
      "unit": "kg",
      "price": -1,
      "type": "product",
      "product_type": "fruit_and_vegetable",
      "tiliter_id": "75fe86a5-2863-4ddc-af7a-73628824b245",
      "score": 12.11
    }
  ]
}
```


It contains a list of 6 items:

- *guid*
- *weight_kg*
- *prediction_time*
- *organic_info*
- *bag_predictions*
- *predictions*

These are explored in detail below.

Note: some fields may not be presented if they're not applicable.

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Key1	Key2	Description	Examples
guid	N/A	Globally unique ID of image used in the prediction. Used to calculate real world model performance.	d6100d1d-4393-4169-bf6f-1b70ecca18c8
weight_kg	N/A	Weight of the item in kilograms or pounds (depending on region and scale used). This is set to *null* if no scale is connected or used in the application.	0.855 = 0.855kg or lb 0.13 = 0.13kg or lb 1.62 = 1.62kg or lb 6.825 = 6.825 kg or lb
prediction_time	N/A	Time in seconds taken to make the prediction.	0.4673443465655 0.3764353443266
organic_info	*accuracy*	Accuracy score for the product being organic. Can only be values from 0-100.	80 99.73 1.4
	threshold	Threshold Prediction for organic accuracy. Can only be values from 0-100.	30
bag_predictions	*bag_name*	The top-level bag type name *none* = no bag *bakery* = brown paper bag *net* = reusable net *plastic* = plastic bag	
	score	Prediction score for bag type. Can only be values from 0-100.	80 99.73 1.4
predictions	*category*	The top-level category of fresh produce item.	Apple (for Apple – Granny Smith)
	variety_1	First level of variety.	Truss (for Tomato – Truss – Cherry)
	variety_2	Second level of variety.	Cherry (for Tomato – Truss – Cherry)
	full_name	The full name of fresh produce item (includes variety1 and variety2).	Apple – Granny Smith
	custom_name	Custom Name identifier provided by customer.	Granny Smith Apple
	custom_id	Custom ID representing that product in the retailer's internal database. Can be used to map to the retailer's database.	4017

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	custom_id_organic	Custom ID of the organic variety of this item. Can be used to map to the retailer's database, but only if the *organic_info* value exceeds the threshold, and the *custom_id_organic* has a non-zero value.	94017
	unit	Unit the produce is sold by.	kg ea
	type	Response type.	Product No fresh produce No response Hand Low score
	product_type	The product type, used to broadly categorise the results. In this case it is fruit_and_vegetable.	fruit_and_vegetable nut dried_fruit bakery snack special missing
	tiliter_id	A globally unique ID corresponding to this particular category. Can be used to map to the retailer's database.	21324b9e-a35c-48d1-89de-c3036607422a
	score	Prediction score 0-100 e.g., 80 = 80%	80 99.73 1.4

There's a lot of information in this file, but it's critical to extract the results from the *predictions* field, which the POS unit can read and map them to internal items.
The lookup key for this mapping can be either the *tiliter_id* or the *custom_id*. These need to be mapped to the retailer's items for accounting purposes and to charge the correct amount. See the How to manage active categories section below for more details.

These should be extracted and stored as a list, along with the retailer's internal identifier and the corresponding *score*. And this list should be passed to the next stage for developing the UI.

Generally, all predictions contained in the JSON response should be included in this list. If the JSON response contains a category, regardless of the score or any other factor, it's considered to be a meaningful prediction by the Tiliter Vision processor.

There's no need for the JSON list to be further filtered based on *score as* the JSON response only includes predictions that exceed a pre-defined threshold. This is calculated at model training time and is dependent on the category and the rank of the item in the list of predictions.

Only categories that are set to active in the **activelist** (see *How to manage active categories* below for more details) are included in the JSON response. There are some special case items that are not produce and should be filtered out. These are explained below.

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Special cases

The Tiliter Vision system is trained to recognise various non-produce items. If these get the highest rank score in the prediction response, the retailer can decide to take special action to ensure they are correctly handled. These decisions should be made during project implementation. These cases are described below.

Background

When a prediction is triggered with an empty or near empty scale, a prediction of **background** will be made. An example of this is below. This category should never be shown as a selection to the shopper but should be handled in the POS unit's logic instead.

The **background** prediction can be used to detect a zero error – see Zero error detection below for more information.


```
{
  "guid": "650861f9-3d2f-437c-a416-a33d440425d5",
  "weight_kg": null,
  "prediction_time": 0.20077276229858398,
  "bag_predictions": [
    {
      "bag_name": "none",
      "score": 100
    },
    {
      "bag_name": "bakery",
      "score": 0
    },
    {
      "bag_name": "net",
      "score": 0
    },
    {
      "bag_name": "plastic",
      "score": 0
    }
  ],
  "organic_info": {
    "accuracy": 2.76,
    "threshold": 30
  },
  "predictions": [
    {
      "category": "Background",
      "variety_1": null,
      "variety_2": null,
      "full_name": "Background",
      "custom_name": "0",
      "custom_id": "0",
      "custom_id_organic": "0",
      "unit": "kg",
      "price": -1,
      "type": "product",
      "product_type": "special",
      "tiliter_id": "d9e7af35-5946-4ad5-8f31-505cf051e478",
      "score": 100
    }
  ]
}
```


Non-produce – Hand

If the Tiliter Vision system detects a hand or part of a hand, it will return **Non-produce – Hand**.

This detection can be set up to initiate an automatic retry, or to present a warning to the shopper asking them to remove their hand. See Auto-retry on hand detection below for more detail.


```
{
  "guid": "8b4cd2b4-0f96-40ad-ba3e-50e1ac4f0415",
  "weight_kg": 0.34,
  "prediction_time": 0.356749153137207,
```


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```
  "bag_predictions": [
    {
      "bag_name": "none",
      "score": 100
    },
    {
      "bag_name": "bakery",
      "score": 0
    },
    {
      "bag_name": "net",
      "score": 0
    },
    {
      "bag_name": "plastic",
      "score": 0
    }
  ],
  "organic_info": {
    "accuracy": 0.49,
    "threshold": 30
  },
  "predictions": [
    {
      "category": "Non",
      "variety_1": "Produce",
      "variety_2": "Hand",
      "full_name": "Non-Produce - Hand",
      "custom_name": "0",
      "custom_id": "0",
      "custom_id_organic": "0",
      "unit": "kg",
      "price": -1,
      "type": "product",
      "product_type": "special",
      "tiliter_id": "c263cd0f-a3e3-4aef-845e-e3d361fba47b",
      "score": 99.22
    }
  ]
}
```


Low score

If the Tiliter Vision system detects something that is not a produce item, such as a box of cereal or a bottle of water, it will return **Low Score**.


```
{
  "guid": "f191095e-b3db-4dd8-a0a5-03ef915d71a9",
  "weight_kg": 0.225,
  "prediction_time": 0.6742618083953857,
  "bag_predictions": [],
  "organic_info": {
    "accuracy": 0.94,
    "threshold": 30
  },
  "predictions": [
    {
      "full_name": "Low score",
      "type": "Low score",
      "score": 100,
      "custom_id": "Low score"
    }
  ]
}
```


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Organic items

You can configure Tiliter Vision to recognise organic items based on markings. This requires development work, so please discuss this with your local Tiliter representative.

Once this has been set up, the *organic_info* field will become active and can be recognised by the POS unit. Once the *organic_info* field *accuracy* exceeds *threshold* then the item is considered to be an organic item.

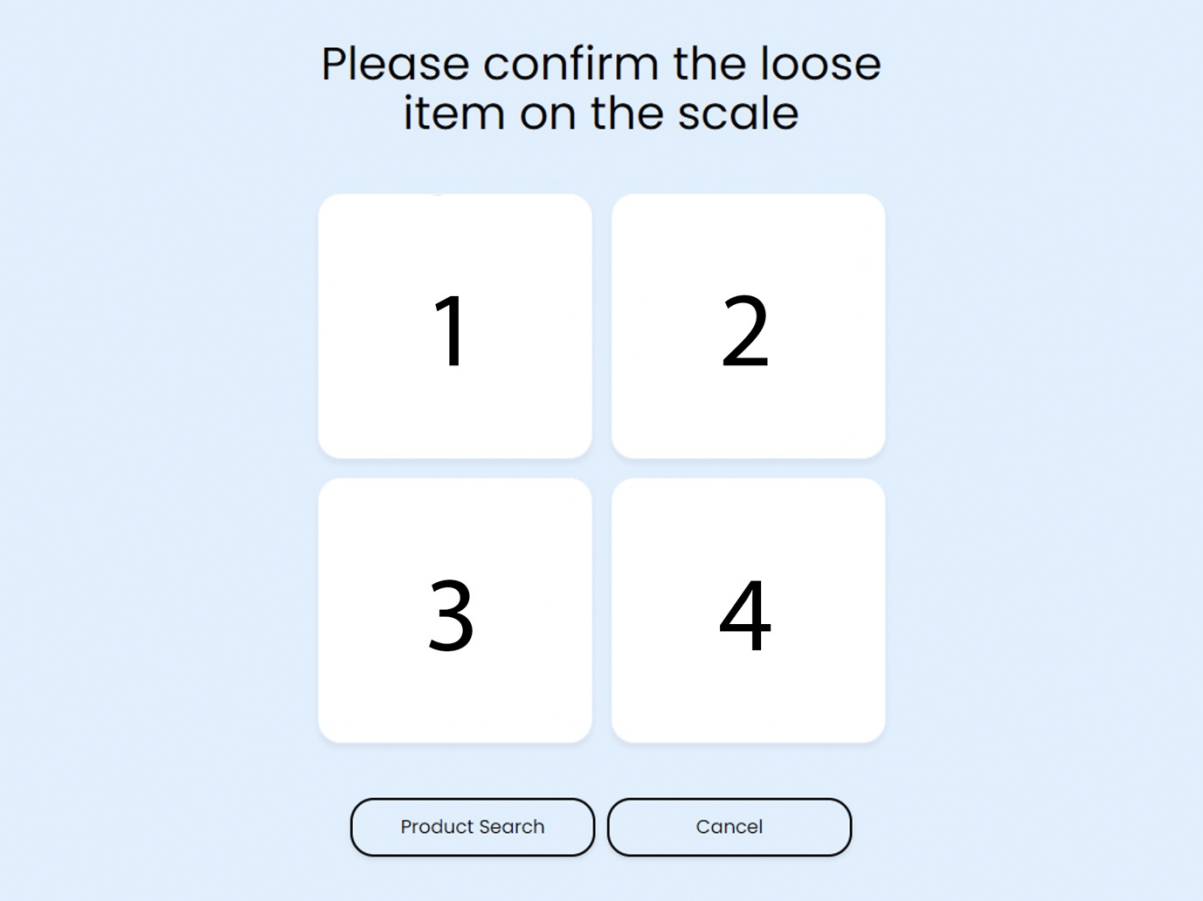
In this case, the *custom_id_organic* can be used as the lookup key. Alternatively, the *tiliter_id* can still be used as long as the POS unit contains logic to map to the corresponding organic item in the retailer's internal produce list.

Options are presented to the shopper on the screen

Once a prediction has been retrieved and interpreted by the POS unit, it's presented to the shopper on the screen.

The input for the screen is the ordered list of items from the JSON response, with any special case categories filtered out. Options should be presented to the shopper in priority order based on their rank. Here's our suggestion for the positioning:

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The shopper makes a selection

One category only

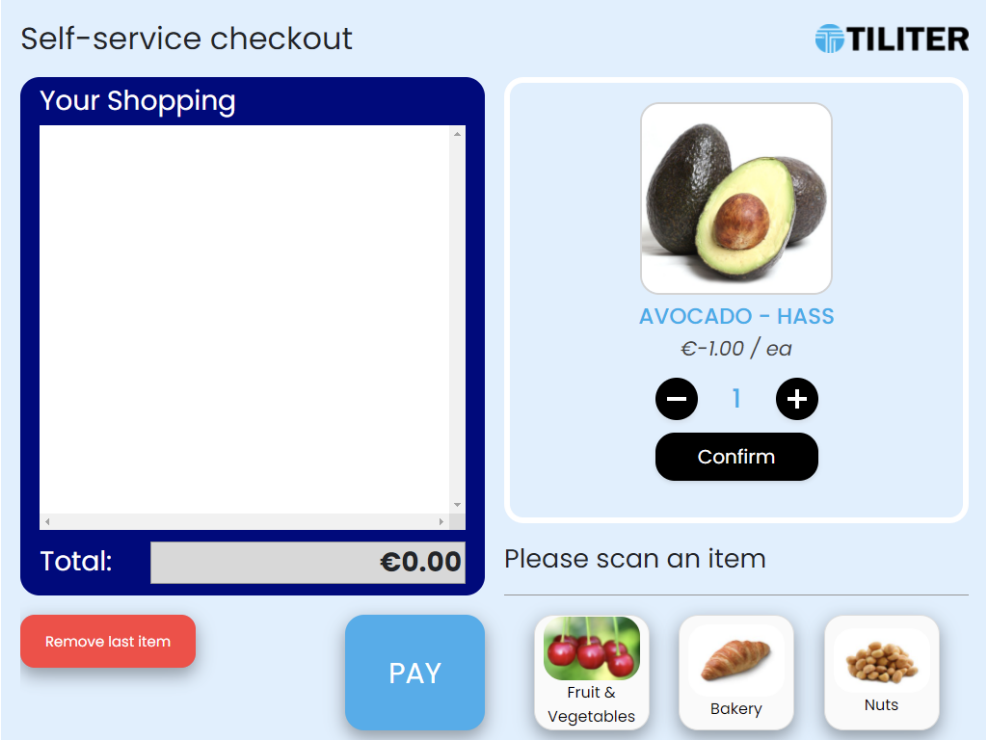
When only one category is predicted, this can be automatically added to the cart. This comes with a risk if it's difficult for the shopper to remove an incorrect item.

The retailer can decide whether the item would be added automatically, or a selection (of one) is still needed.

Sold by quantity

Items that are sold by quantity must present the shopper with a quantity selection screen. The screen can default to a quantity based on dividing the actual weight by the average weight as specified in the **activelist**. This can limit accuracy, especially when the quantity increases. So, the shopper should always be able to override this.

Here's an example of how you can present the shopper with a quantity selection screen:



Limiting the number of results

If the retailer would like a maximum number of options presented to the shopper, this should be configured by filtering items based on a rank method, not based on score using a threshold method.

Here's how to do this:

1. The retailer decides the maximum number of choices to show on screen, for example: 4.
2. The system reads the results.
3. If there are 4 or less results, the system will show them all.
4. If there are more than 4, the system will filter ones with the lowest score, so the total list is 4. For example, if we show 10, filter out the lowest 6, and only present the remaining 4.

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Note: The score can be extremely low but could still be relevant.

In this example, **all three items should be presented**, despite Apple - Jazz only being given a score of 0.001.

1. Apple – Pink Lady: 0.9
2. Apply – Royal Gala: 0.09
3. Apple – Jazz: 0.001

In this example, the POS should **truncate the categories with the lowest two scores**, i.e., Apple – Red Delicious and Apple – Envy despite being given a score of 0.1 (greater than Apple – Jazz above).

1. Apple – Pink Lady: 0.2
2. Apply – Royal Gala: 0.2
3. Apple – Jazz: 0.2
4. Apple – Kanzi: 0.2
5. Apple – Red Delicious: 0.1
6. Apple – Envy: 0.1

The shopper's selection is posted to the Tiliter Vision unit

Sometimes the system will offer several prediction options — for example, Apple – Pink Lady and Apple – Royal Gala. In this case, the shopper needs to choose between the two options.

At other times, the shopper may completely override the system, either because the image is blocked by a bag, or the shopper has made a mistake. To take advantage of Tiliter's theft prevention feature and to get better analytics, we need to know which selection the shopper chose.

Here's the workflow:

1. The shopper places the item on the scale.
2. The shopper makes their selection on screen.
3. The POS PC posts the selection to the label_last_result endpoint

And here's how to post to the endpoint:

1. After the shopper makes a selection, the POS PC must make an HTTP POST request in this format:

curl --Location --request POST 'http://DEVICE IP OR **HOSTNAME:8000/backend/label_last_result'**

How to manage active categories

The Tiliter Vision system will only include categories in the JSON response if they're active. This prevents incorrect results, such as items that aren't found in that store, from being included and results in improved performance.

There are several configurable fields per category. Most important is *custom_id* which is how we translate the Tiliter category into the article ID used by the retailer. It's also important to set the unit to either kg or ea, to show that the item is sold by weight or quantity.

There are several ways to manage the **activelist**, described below. For more information, see the **Active Category Management** document or speak to your local Tiliter representative.

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Option 1: Automatic integration

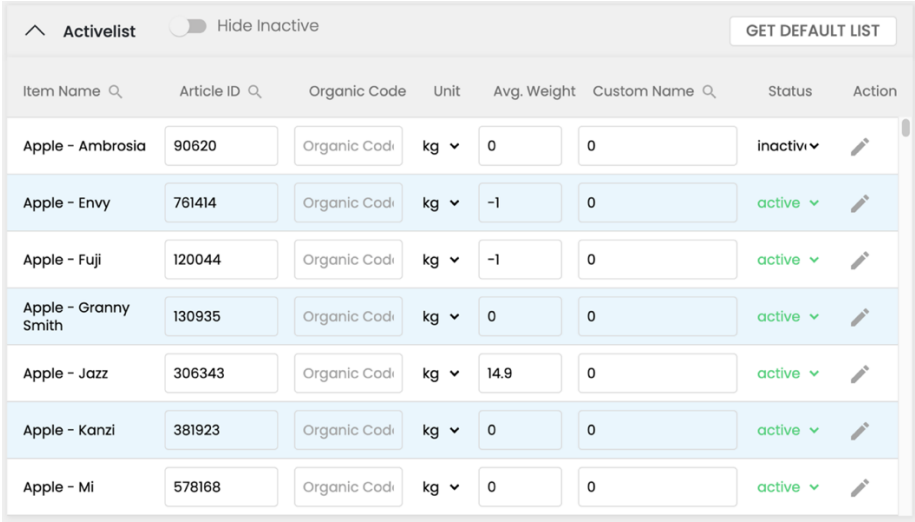
The best way to manage active categories is to integrate an existing produce list with the Tiliter backend. This needs to be set up by Tiliter and the retailer, and once it's complete, the Tiliter Vision system will automatically synchronise with the produce list.

Please speak to one of our friendly service team at service@tiliter.com for more information on getting this setup.

Option 2: Manual management via Tiliter Portal

You can manually manage all devices in a store at once, using the Tiliter Portal. The device will need to be connected to the internet to do this. Refer to the **Network Access Requirements** document for more information or speak to your friendly local Tiliter representative.

The Tiliter Portal can be accessed at https://tiliter.portal.com, and you can manage the **activelist** by navigating to Devices > Deployment > Device Configuration.



Option 3: Manual management on the device

If you can't connect the device to the internet, you can manage the **activelist** via this endpoint:

http://<DEVICE IP ADDRESS OR HOSTNAME>:8000/activelist

Examples:

- For software running on an external device with IP Address 169.254.240.146 the address is http://169.254.240.146:8000/activelist/
- For software running on an external device with hostname TV123, the address is http://TV123:8000/activelist/
- For software running on the host machine the address is http://localhost:8000/activelist/

Note: once the device has internet access and the portal is configured, the portal list will be the primary list and will override the local list.

This endpoint also provides a user interface screen if you visit via a web browser.

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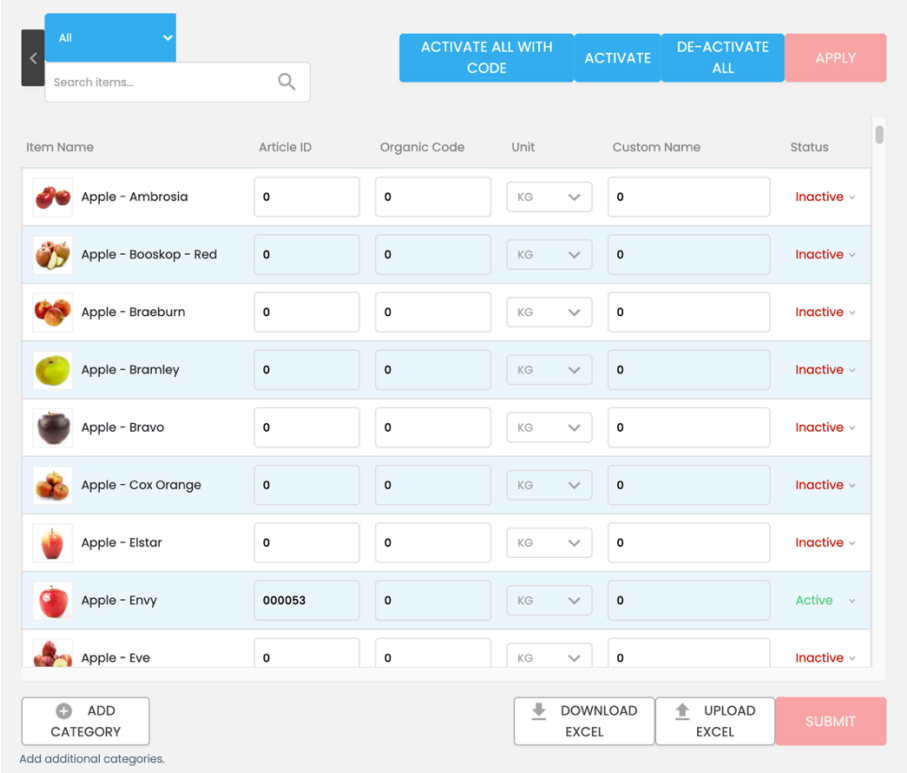
How to change active category status

If a category is no longer active, due to seasonality or supply for example, you should deactivate it. This prevents it from showing in the predictions and keeps performance optimal. To set a category's active/inactive status, simply tick or un-tick the corresponding checkbox.

There are buttons to select categories with a custom id, and you can select/deselect all. Once you've made your changes, press the **Update categories** button to save.

Active categories example

Here's an example of the active categories mapping page (**note:** this example shows an **Apply changes** button — your screen might show **Update categories** or similar).



Advanced features summary

As shown in the UI design examples, there are several advanced features that a POS software integrator can build into the POS unit by taking advantage of the Tiliter Vision recognition. Here's some more detail.

Auto-retry on hand detection

If the image capture contains a part of the shopper's hand, it will be detected and reported. The POS unit can automatically trigger another prediction in this case, to make the experience seamless.

The logic could be that the POS triggers up to 5 predictions in a row, and if a hand still is in the frame, then a message can be shown on the screen.

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Zero error detection

The Tiliter Vision system can recognise the background, so some simple logic can be built into the POS unit to allow the detection of a zero error. To do this, whenever a prediction is triggered that returns **background**, and the scale weight is non-zero, an error is shown onscreen to alert the shopper of a zero error.

This could be taken one step further and the POS could be "soft locked" to prevent new predictions until the scale reads zero again.

Quantity Estimation

Using the weight and average weight, the POS unit can calculate an estimate of the quantity of items on the scale. This can be used as the default quantity value for all items sold by quantity.

Integration Dos and Don'ts

Integration dos and don'ts for POS software providers.

Do:

- Make it very easy and obvious for the shopper to initiate a prediction. Stable weight is preferred, if not then use a single large button with clear, large font. This must be highly intuitive.
- Present the shopper with a list of choices that are ranked by score. There should be large buttons taking up as much of the screen as practical, with preview images. The buttons should be centred on the screen with the item name centred underneath each.
- Present the shopper with an override option that takes them to an easy to navigate search screen. This should have logical first letter groupings, and category groupings (e.g., Apples, Mangos, Mandarins). The buttons should be easy to read and well presented.
- Interpret the JSON field "unit" and if it is "ea" then present the shopper with a selection for quantity.
- Use the "average weight" field combined with the weight reading to make an initial estimate on the quantity.
- Show a friendly message when the top prediction is "hand" - or consider an automated retry.
- Show a "zero error" message when the system predicts background, but the weight is non zero. This can even lock the system from use to prevent mistaken weights.
- Send Tiliter screenshot, videos, or software builds throughout the integration process to ensure alignment.

Don't:

- Show an error up if the Tiliter Vision system doesn't respond. Instead, open the search menu.
- Show an error up if there are no produce items in the Tiliter JSON response. E.g., only background, hand, low score etc. These cases can be handled independently, or the shopper could be taken to the search screen.
- Automatically add an item to the cart if it's difficult to remove. Instead, present a choice between adding to the cart, and opening the search menu.
- Show an error if a barcode for an item sold by weight is scanned. Instead, it should be ignored, or a message should be showing saying "please place the item on the scale."
- Perform all integration tasks in isolation — please contact Tiliter for feedback and assistance anytime.

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